THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 1: **Exploration: Entering the World of Chemistry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student models a falling ball ignoring air resistance. For which object is the model least accurate? 1

(a) an iron ball **(b)** a stone **(c)** a feather **(d)** a steel bolt

2. A pendulum model assumes no friction. The actual pendulum: 1

(a) swings forever **(b)** stops at once
(c) speeds up **(d)** slowly stops swinging

3. The density of an object is 2.5 g per cubic cm. In kg per cubic metre this is: 1

(a) 2500 **(b)** 25000 **(c)** 250 **(d)** 25

4. A runner covers 100 m in 12.5 s. The estimated speed is: 1

(a) 8 m/s **(b)** 80 m/s **(c)** 12.5 m/s **(d)** 1.25 m/s

5. The predictions of Newton's law of gravitation matched the observed motion of planets. This: 1

(a) supports the law **(b)** has no effect
(c) refutes it **(d)** proves it forever

6. 72 km/h in m/s is: 1

(a) 7.2 **(b)** 20 **(c)** 72 **(d)** 200

7. A car travels 90 km in 1.5 hours. Its speed in m/s is: 1

(a) 6 m/s **(b)** 90 m/s **(c)** 16.7 m/s **(d)** 60 m/s

8. The 'ideal gas' model works best at: 1

(a) high pressure and high temperature **(b)** any condition
(c) high pressure and low temperature **(d)** low pressure and high temperature

9. The area of a rectangle 50 cm by 30 cm in square metres is: 1

(a) 1.5 **(b)** 15 **(c)** 0.15 **(d)** 0.015

10. The volume of a cuboid 2 m x 50 cm x 20 cm in cubic metres is: 1

(a) 20 **(b)** 2 **(c)** 0.2 **(d)** 200

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A model assumes frictionless motion. Give one situation where it is acceptable and one where it fails. 2

12. Why do scientists express large numbers in scientific notation? Write 300000000 in this form. 2

13. How is a hypothesis different from a prediction? 2

14. A student takes five readings of a mass: 5.1, 5.0, 5.2, 5.1, 5.1 g. Find the mean. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student measures the time period of a pendulum by timing 20 oscillations as 32 s. Find the time period and say why 20 oscillations are timed. 3

16. Explain the importance of scientific notation, using examples of the size of an atom and the distance to the Sun. 3

17. Explain how a graph helps to identify a mistake in a measurement. 3

OR

The diameter of a hair is 0.07 mm. Express it in metres and in micrometres. 3

18. Explain how an astronomer can find the distance to a star in light years and what a light year is. 3

OR

State three safety rules for chemical laboratories and the reason for each. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A pendulum experiment gave: length (cm) 25, 50, 75, 100 and time for 10 oscillations (s) 10, 14.2, 17.4, 20. Calculate the time periods, comment on the relationship and mention the assumptions. 5

OR

Explain how scientists establish a theory (e.g. cell theory or atomic theory) through evidence over time with at least five steps. 5

20. Explain the use of estimation in science with three worked examples and discuss the sources of uncertainty. 5

OR

Discuss the sources and effects of units confusion in engineering projects such as the aircraft fuel incident. 5

END OF QUESTION PAPER — Total: 40 Marks

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